

SAUMYA SINHA

Data Analyst

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LinkedIn

Seeking a challenging and dynamic opportunity to leverage advanced data analytics, cloud-based data engineering, and visualization expertise to drive actionable insights, optimize business processes, and support data-driven decision-making. Committed to transforming complex datasets into strategic intelligence through innovative ETL solutions, predictive modeling, and interactive dashboards that deliver measurable business impact.

Education

- 2023:** M.S. Data Analytics, San Jose State University
San Jose, CA, USA
- 2016:** B.Tech., Computer Science, Biju Patnaik University of Technology
Bhubaneswar, Odisha

Core Competencies

Data Analytics & Business Intelligence

ETL Pipeline Development

SQL & Relational Database Optimization

Python Statistical Modeling

AWS Cloud Data Engineering

Dashboard Development

Exploratory Data Analysis (EDA)

Predictive & Prescriptive Analytics

A/B Testing & Hypothesis Testing

Big Data Processing (PySpark / AWS Glue)

Profile Summary

- Gained nearly 5+ years of experience in advanced data analytics, statistical modeling and predictive insights to transform complex datasets into actionable business intelligence, driving operational efficiency and strategic decision-making.
- Previously worked as Data Analyst VIVIO Health Inc., leveraged advanced analytics, Python, SQL, and AWS cloud technologies to automate ETL pipelines, develop interactive dashboards, and deliver actionable insights that drive operational efficiency and strategic decision-making.
- Expertise in automated high-volume reporting and ETL workflow using Python, SQL, AWS Glue, and PySpark, reducing manual effort and accelerating data availability for business stakeholders.
- Proven ability to design and deploy interactive dashboards in Tableau, Power BI, and QuickSight, enabling real-time monitoring of KPIs, cost trends, and operational performance.
- Skilled in data engineering and cloud technologies (AWS S3, Redshift, SageMaker, DynamoDB), optimizing storage, retrieval, and processing of large-scale datasets for analytics and predictive modeling.
- Leveraged strong problem-solving skills, effective communication & exceptional time management to deliver accurate, high-quality data insights and solutions within tight deadlines.

Work Experience

Data Analyst | VIVIO Health Inc. | Hayward, California | Sep 2024 – Mar 2025

- Spearheaded automation of high-volume financial reporting leveraging MySQL, Python, Excel, RunDeck, and Cursor AI, eliminating 40% of manual effort while achieving flawless data accuracy.
- Engineered and deployed automated claims reporting frameworks using SQL and Amazon QuickSight, uncovering 20–30% potential savings on specialty drugs and proactively flagging anomalous treatment patterns.
- Optimized operational efficiency by 30% through end-to-end automation of specialty drug list monitoring pipelines, mitigating drug fill denials and enhancing pharmacy-patient coordination.
- Applied advanced statistical analytics including hypothesis testing, regression, and correlation modeling in Python to extract actionable cost-saving insights and streamline drug spend optimization.
- Directed end-to-end reporting initiatives from requirements elicitation, data modeling, and dashboard development to stakeholder handoff, consistently delivering 50+ ad hoc extracts and compliance reports with measurable business impact.
- Used Excel for data checks and audit reporting, leveraging Pivot Tables, VLOOKUP/XLOOKUP, conditional formatting, and functions to track variances, validate claims, and summarize findings.

Data Analyst | San Jose State University | San Jose | Feb 2024 – Sep 2024

- Engineered intelligent data extraction pipelines by leveraging LayoutLMv3 and PyPDF to parse key-value pairs from PDF forms, storing structured outputs in AWS S3 for seamless downstream processing.
- Accelerated data transformation and ingestion by designing PySpark ETL pipelines in AWS Glue, optimizing large datasets for high-performance querying in AWS Redshift, enhancing analytical efficiency by 25%.
- Developed interactive Tableau dashboards and comprehensive workflow documentation to visualize approval trends and demographic insights, contributing to a 15% increase in program acceptance rates.



Technical Skills

- **Data Analysis & Statistical Modeling:** Exploratory Data Analysis (EDA), Data Visualization, Data Cleaning & Transformation, Data Modeling, Statistical Modeling, A/B Testing & Hypothesis Testing
- **Programming & Scripting:** Python (Pandas, NumPy, SciPy, Statsmodels), R, Java, JavaScript
- **Data Engineering & ETL:** SQL: MySQL, SQL Server, ETL Pipelines, AWS Glue, Apache Airflow
- **Data Visualization & Reporting:** Tableau, Power BI, Amazon QuickSight, Google Analytics, Matplotlib, Seaborn, Plotly, Excel (Pivot Tables, VLOOKUP, HLOOKUP)
- **Cloud Computing & Big Data:** AWS S3, AWS Redshift, AWS SageMaker, DynamoDB, PySpark



Certifications

- [Google Data Analytics Professional Certificate](#)
- [Amazon Redshift](#)
- [MS Excel](#)
- [Amazon SageMaker](#)
- [Tableau](#)

- Implemented best practices in **data validation, pipeline automation, and metadata management**, improving overall data quality and reliability for research initiatives.

Analyst | Optum Inc. / UnitedHealth Group | Hyderabad | May 2019 – Dec 2021

- Boosted claim processing efficiency by **20%** and reduced operational costs by **15%** by defining KPIs, performing end-to-end analysis of patient claims using **SQL and Python (Pandas, Statsmodels)**, and streamlining workflows.
- Delivered **strategic insights** through **AWS Redshift-powered Tableau dashboards**, enabling senior management to monitor cost trends and claim approval rates, realizing **5% operational cost savings**.
- Led statistical analyses of **A/B testing outcomes**, employing Python for modeling and Tableau for visualization, resulting in a **10% improvement in product adoption**.
- Standardized reporting frameworks and data governance practices, ensuring accuracy, compliance, and actionable insights across multiple business units.

Member Technical | ADP (Automatic Data Processing) | Hyderabad | Sep 2016 – May 2019

- Collaborated with business and operations teams to define KPIs and measure business outcomes, performing **comprehensive EDA on HCM and payroll datasets** using **MySQL and Python** to support ETL workflows.
- Executed **data cleaning, advanced analytics, and statistical modeling** with Python libraries (Pandas, SciPy, Statsmodels), generating actionable insights and improving decision-making accuracy.
- Designed and deployed **20+ interactive dashboards and reports** using Tableau, Excel, Matplotlib, and Seaborn, enabling **real-time monitoring of key performance metrics** tailored to business objectives.
- Championed data quality initiatives, automated reporting processes, and enhanced operational transparency, driving efficiency and stakeholder confidence.